Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Unitary evolution as infinitesimal projective processes

Files.

- File: `briefs/UnitaryProjectiveEvolution/UnitaryProjectiveEvolution.lean`
- Library: `UnitaryProjectiveEvolution`
- Imports:
  `ForPhyslib.QuantumMechanics.VonNeumannResolvent`,
  `ForPhyslib.QuantumMechanics.Wigner.AffineExtension`,
  `ForMathlib.Analysis.SpecialFunctions.Exponential`,
  and the brief `SchroedingerEvolution` (for `IsEvolution`, `DrUnit`, `DrEv`, `DrEv.drUnit` and Proposition 1)
- New file `ForPhyslib/QuantumMechanics/VonNeumannResolvent.lean`: the von Neumann generator, the unitary channel of physlib's `UnitaryOneParameterGroup`, the resolvent and its spectral identity, the second-order estimates, the largest outcome on a pure state with a Davis–Kahan bound, the resolvent exponential formula and time reversal. `Pinching` gained the spectral decomposition `sum_smul_spectralProj` and `spectralProj_ne_zero`.

- New `ForMathlib` material:
  `ForMathlib/Analysis/InnerProductSpace/Trace.lean` (the operator norm of a positive operator is at most its trace).

- For the remarks "The qubit picture" and "The finite process is not unitary", and for the thermodynamic analogy, not imported by the brief's file:
  `ForPhyslib/QuantumMechanics/VonNeumannResolvent/Cayley.lean` (the Cayley transform of L and its Hilbert–Schmidt unitarity),
  `ForPhyslib/QuantumMechanics/Qubit/VonNeumannCayley.lean` (the resolvent of a qubit on Bloch vectors),
  `ForMathlib/LinearAlgebra/Matrix/Cayley.lean` (the Cayley transform of a skew-symmetric matrix is special orthogonal),
  `ForPhyslib/QuantumMechanics/VonNeumannResolvent/Irreversible.lean` (purity decreases, and the inverse $1 - \tau L$ is not positive),
  `ForPhyslib/QuantumMechanics/VonNeumannResolvent/QuasiStatic.lean` (purity loss and entropy production vanish in the quasi-static limit),
  `ForMathlib/Analysis/SpecialFunctions/Pow/NegMulLog.lean` (the Shannon entropy is at most $8n(1 - \sum_i p_i^2)^{3/4}$).

Setting. E is a finite-dimensional complex Hilbert space, with density operators, pure states and `AffineOnDensities` as in the brief "Multiple equivalent characterizations of the Schrödinger equation". V is a norm-continuous unitary one-parameter group with self-adjoint generator K, $V_t = e^{-itK}$; $\hbar = 1$, so $K = H/\hbar$. `vonNeumannGenerator` $K\ \rho = -i(K\rho - \rho K)$ is L, `unitaryChannel` $V\ t\ \rho = V_t \rho V_t^\dagger$ is T_t, and `vonNeumannResolvent` $V\ \tau = (1 - \tau L)^{-1}$ is Ψ_τ, a linear equivalence of $\mathrm{End}(E)$. The definitions are

- `IsProjectiveDecomposition` P: star projections P_a, $P_a P_b = 0$ for $a \neq b$, $\sum_a P_a = 1$, over a finite index type;

- `IsStateAdaptiveProjective` Ψ: Ψ maps density operators to density operators, is affine on them, and for each density operator ρ there is a projective decomposition with $\Psi\rho = \sum_a P_a \rho P_a$ (Definition 1);

- `IsOutcome` $\Psi\ P\ p\ Q$: Q is pure and $F_a P F_a = pQ$ for one term of a projective decomposition with $\Psi P = \sum_a F_a P F_a$; `IsBranch` $\Psi\ n\ Q\ p$: Q_0 is pure and each Q_{k+1} is an outcome on Q_k of probability p_k, for $k < n$;

- `PerfectPrediction` $\Psi\ t$: from every pure P there are, for all large n, branches $Q^{(n)}$ of n steps of $\Psi_{t/n}$ with $Q_0^{(n)} = P$, whose probability $\prod_k p_k^{(n)}$ tends to 1 and whose final state $Q_n^{(n)}$ converges;

- `IsProjectiveLimit` $\Phi\ \Psi$: every Ψ_τ with $\tau > 0$ is state-adaptive projective, $(\Psi_\tau \rho - \rho)/\tau$ converges as $\tau \to 0^+$ for every density operator, $(\Psi_{t/n})^n \rho \to \Phi_t \rho$ for $t > 0$, and prediction is perfect for every $t > 0$;

- `DrProj` Φ (DR-PROJ): Φ is a projective limit, and a second projective limit Φ' (reconstruction) satisfies $\Phi'_t(\Phi_t P) = P$ and $\Phi_t(\Phi'_t P) = P$ for pure P and $t > 0$;

- `DrUnit` (DR-UNIT) and `IsEvolution` are those of the brief on the Schrödinger equation.

Results.

Brief	Lean
Definition 1	`IsStateAdaptiveProjective`
Proposition 2, $(I - \tau L)^{-1}$ exists	`injective_vonNeumannStep`, `vonNeumannResolvent`
Eq. (10)	`vonNeumannResolvent_sub`
Eq. (11)	`spectralProj_mul_vonNeumannGenerator_mul`
Eq. (12)	`spectralProj_mul_mul_vonNeumannResolvent`
Eq. (4), $\Psi_\tau \rho = \sum_a P_a^\rho \rho P_a^\rho$	`pinching_vonNeumannResolvent`
Ψ_τ is affine, positive, trace preserving	`isDensity_vonNeumannResolvent`, `trace_vonNeumannResolvent`
Eq. (7), and Ψ_τ is completely positive	`vonNeumannResolvent_eq_laplaceResolvent`, `laplaceResolventCP` (any-dimension part)
Proposition 2	`isStateAdaptiveProjective_vonNeumannResolvent`
$\|L_K\| \leq 2M/\hbar$	`norm_vonNeumannGenerator_apply_le`
Eq. (16)	`norm_vonNeumannResolvent_sub_sub_le`, `norm_unitaryChannel_sub_sub_le`
Proposition 13, eq. (14)	`norm_vonNeumannResolvent_sub_unitaryChannel_le`: $\leq 8\|K\|^2\tau^2$
Proposition 13, eq. (15), and $F_\tau(P)$ has rank one	`exists_spectralProj_max_eq_proj`: $1 - q_\tau(P) \leq 8\|K\|^2\tau^2$, $\|F_\tau(P) - T_\tau P\| \leq 48\|K\|^2\tau^2$
Proposition 17	`implemented_of_bijOnPure`
$(\Psi_\tau - I)/\tau = L\Psi_\tau \to L$	`tendsto_inv_smul_vonNeumannResolvent_sub`
Eq. (20)	`tendsto_vonNeumannResolvent_iterate`, error $\leq 8\|K\|^2 t^2/n$
Eq. (21), errors accumulate to $O(t^2/n)$	`perfectPrediction_vonNeumannResolvent`, `norm_sub_unitaryChannel_le_of_step`
"Replacing H with $-H$"	`timeReversal`, `unitaryChannel_timeReversal`
DR-UNIT ⇒ DR-PROJ	`DrUnit.drProj`
The branch has weight tending to one in the unconditioned output	`isPositive_iterate_sub_of_isBranch`, `norm_iterate_sub_le_of_isBranch`
Φ_t maps pure states to pure states	`IsProjectiveLimit.isPure`, `isClosed_setOf_isPure`
Φ_t maps pure states bijectively onto pure states	`DrProj.drEv`
Eq. (27), phases	from the brief on the Schrödinger equation, `DrEv.drUnit`
DR-PROJ ⇒ DR-UNIT	`DrProj.drUnit`
Theorem 18	`drUnit_iff_drProj`
A map affine on density operators is linear on them	`exists_basis_isDensity`, `AffineOnDensities.basis_constr_apply`, `AffineOnDensities.exists_linearMap`
$(T_{t/n})^n \to e^{tB}$ when $(T_\tau - I)/\tau \to B$	`tendsto_pow_exp_of_tendsto_smul_sub_one`
Affinity of the limit, eqs. (28)–(29), composition for all durations	`IsProjectiveLimit.exists_eq_exp` ($\Phi_t = e^{tB}$ on density operators), `IsProjectiveLimit.isEvolution`, `DrProj.isEvolution`

Brief	Lean
Theorem 18 without `IsEvolution`	`drUnit_iff_drProj_and_map_zero`
Sec. 5, $\Psi_\tau = (I + \mathcal{C}_\tau)/2$	`vonNeumannCayley`, `vonNeumannCayley_vonNeumannStep`, `vonNeumannResolvent_eq_midpoint`
Sec. 5, L skew and $\mathcal{C}_\tau$ orthogonal for the Hilbert–Schmidt inner product	`trace_star_vonNeumannGenerator_mul`, `trace_star_vonNeumannCayley_mul`
Sec. 5, on the Bloch ball $\mathcal{C}_\tau$ is a rotation and Ψ_τ takes the midpoint	`exists_vonNeumannCayley_blochOperator`, `vonNeumannGenerator_blochOperator`, `exists_blochOperator_eq`, `cayley_mem_specialOrthogonalGroup`
Sec. 5, Ψ_τ maps pure ensembles to mixed ones	`re_trace_star_mul_self_eq_vonNeumannResolvent`, `re_trace_star_mul_self_vonNeumannResolvent_lt`, `not_isPure_vonNeumannResolvent`
Sec. 5, $I - \tau L$ is not generally positive	`not_nonneg_vonNeumannStep_proj`, `vonNeumannGenerator_proj_ne_zero`
Sec. 5, "generally": unless K is a multiple of the identity	`exists_isPure_not_isPure_vonNeumannResolvent`, `forall_isPure_vonNeumannResolvent_iff`, `vonNeumannResolvent_eq_self_of_eq_smul`
Secs. 1 and 6, thermodynamic analogy: purity loss per step $O(\tau^2)$, in total $O(1/n)$	`hsNormSq_vonNeumannGenerator_le`, `hsNormSq_sub_hsNormSq_vonNeumannResolvent_le`, `hsNormSq_sub_hsNormSq_vonNeumannResolvent_iterate_le`, `tendsto_hsNormSq_sub_hsNormSq_vonNeumannResolvent_iterate`, `tendsto_sum_one_sub_hsNormSq_vonNeumannResolvent_proj`
Secs. 1 and 6, thermodynamic analogy: entropy production per step $O(\tau^{3/2})$, in total $O(1/\sqrt{n})$	`Wigner.IsDensity.vonNeumannEntropy_le`, `sum_negMulLog_le`, `vonNeumannEntropy_vonNeumannResolvent_proj_le`, `tendsto_sum_vonNeumannEntropy_vonNeumannResolvent_proj`

Differences from the brief.

- Finite dimension only; the brief has a separable Hilbert space. Every generator is bounded, so the cutoffs $H_\tau = f_\tau(H)$, eqs. (22)–(25) and the remark on unbounded generators are not needed here; Sec. 1.1 formalizes them.
- Norms are operator norms, not trace norms. Density operators have operator norm at most one, so the bounds of Proposition 13 are uniform over density operators as in the brief, with the explicit constants 8 and 48. In finite dimension trace-norm convergence follows, but a trace-norm version of (14) and (15) would carry the dimension.
- Positivity and trace preservation of Ψ_τ come from eq. (12), not from the integral representation (7): $\Psi_\tau\rho = \sum_x P_x \rho P_x$, which is a density operator when ρ is. The same identity gives injectivity of $1 - \tau L$, so Proposition 2 holds for every real τ. Eqs. (5)–(6) are not formalized. Eqs. (7)–(9) and complete positivity come from the any-dimension part: there Ψ_τ is the integral (7), and in finite dimension it equals $(1 - \tau L)^{-1}$ (`vonNeumannResolvent_eq_laplaceResolvent`).

- Eq. (15) is proved for the largest eigenvalue, once $8\|K\|^2\tau^2 \le \frac{1}{2}$; it is the only case the proof of Theorem 18 uses.

- DR-PROJ is made precise as follows. Outcomes and branches refer to a projective decomposition that realizes Ψ on the current pure state. Perfect prediction asks that the final branch state converges, without naming the limit; DR-PROJ $\Rightarrow$ DR-UNIT shows that it is $\Phi_t P$. Reconstruction is a second projective limit that inverts Φ_t on pure states. For the resolvents it is the limit of $(1+\tau L)^{-1}$, as in the remark "The finite process is not unitary".

- `drUnit_iff_drProj` assumes `IsEvolution` of the brief on the Schrödinger equation: Φ_t is affine and preserves density operators, $\Phi_0 = I$, $\Phi_{s+t} = \Phi_s\Phi_t$, and $t \mapsto \Phi_t\rho$ is continuous at 0^+. The brief derives these from DR-PROJ (affinity of the limit, eqs. (28)–(29), common refinements). `IsProjectiveLimit.isEvolution` derives them too, but not by the brief's route. Each Ψ_τ agrees on density operators with a linear map T_τ, because the density operators contain a basis of the operators; $(T_\tau - I)/\tau \to B$ in operator norm, because convergence on a basis is norm convergence; and $(T_{t/n})^n \to e^{tB}$ by telescoping. So $\Phi_t = e^{tB}$ on density operators for every $t > 0$, which gives affinity, preservation of density operators, continuity at 0^+ and composition for all durations at once; eq. (28) and the common refinements are not needed. DR-PROJ says nothing about Φ_0, which is assumed to be the identity: `drUnit_iff_drProj_and_map_zero` is DR-UNIT $\Leftrightarrow$ DR-PROJ with $\Phi_0 = I$.

- The qubit picture. $\mathcal{C}_\tau = (1+\tau L)\Psi_\tau$, which equals $2\Psi_\tau - I$ by the resolvent identity, so $\Psi_\tau = (I + \mathcal{C}_\tau)/2$ holds for every real τ. Orthogonality is proved in the stronger complex form $\operatorname{tr}((\mathcal{C}_\tau A)^\dagger \mathcal{C}_\tau B) = \operatorname{tr}(A^\dagger B)$ for all operators: $\mathcal{C}_\tau$ is unitary for the complex Hilbert–Schmidt inner product, hence orthogonal on the real space of self-adjoint operators. For a qubit with $K = c + \frac{1}{2} n\cdot\sigma$, L acts on the Bloch vector a of $\rho_a = \frac{1}{2}(1 + a\cdot\sigma)$ as $a \mapsto n \times a$, a skew-symmetric matrix $[n]_\times$. So $\mathcal{C}_\tau$ acts as the Cayley transform $R = (1+\tau[n]_\times)(1-\tau[n]_\times)^{-1} \in SO(3)$, which fixes n, and $\Psi_\tau\rho_a = \rho_b$ with b the midpoint of a and Ra. The statement holds for every $a \in \mathbb{R}^3$, and every self-adjoint operator of trace one is some ρ_a; that the density operators are exactly the ρ_a with $|a| \le 1$ is not formalized, and the midpoint picture does not need it. The rotation angle, $2\arctan(\tau|n|)$, is not formalized.

- The finite process is not unitary. "Generally" is read as: $\tau \neq 0$ and K is not a multiple of the identity. If K is one, $L = 0$ and $\Psi_\tau = I$; otherwise some unit vector ψ is not an eigenvector of K, and then $L\,|\psi\rangle\langle\psi| \neq 0$. "Mixed" is read as "not pure", through the purity $\operatorname{tr}\rho^2$: with $\sigma = \Psi_\tau\rho$, the skewness of L gives $\operatorname{tr}(\rho^\dagger\rho) = \operatorname{tr}(\sigma^\dagger\sigma) + \tau^2 \operatorname{tr}((L\sigma)^\dagger L\sigma)$, so the purity strictly decreases unless $L\rho = 0$, and a pure state has purity one. "Irreversible" is read as "not undone by a positive map": Ψ_τ is a linear bijection with inverse $I - \tau L$, and $(I-\tau L)\,|\psi\rangle\langle\psi|$ is not positive. For $u = K\psi - \langle\psi|K\psi\rangle\,\psi \neq 0$ and $v = \psi - i(\tau\|u\|^2)^{-1}u$ its expectation value in v is -1. The proofs use neither the Cayley transform nor the spectral theorem. That reconstruction uses $(I+\tau L)^{-1}$, and that only the limit is unitary, is the reconstruction clause of DR-PROJ and Theorem 18 (`drUnit_iff_drProj`).

- The thermodynamic analogy. The brief states no result linking the two limits; the statements formalized are our reading of it. In thermodynamics a reversible quasi-static process is a limit of irreversible equilibration steps whose entropy production is quadratic in the step, so that the total vanishes. Here each step Ψ_τ is irreversible, with a purity loss that is positive unless $L\rho = 0$ (previous item) and equals $\tau^2\|L\Psi_\tau\rho\|_2^2 \le 4\|K\|^2\tau^2\|\rho\|_2^2$, with $\|A\|_2^2 = \operatorname{tr}(A^\dagger A)$ the Hilbert–Schmidt norm. Over n steps of duration t/n the purity loss of $(\Psi_{t/n})^n\rho$ is at most $4\|K\|^2t^2\|\rho\|_2^2/n$, and along any sequence of pure states, as along a branch, the summed loss is

at most $4\|K\|^2t^2/n$. For the von Neumann entropy, $S(\sigma) \le 8d(1-\operatorname{tr}\sigma^2)^{3/4}$ in dimension d, from $-x\log x \le 8(x(1-x))^{3/4}$ on $[0,1]$. So one step on a pure state produces entropy at most $8d\,a\sqrt{a}$ with $a = 2\|K\||\tau|$, and the entropy summed along any sequence of pure states is at most $8d\,c\sqrt{c/n}$ with $c = 2\|K\||t|$. The bound carries the dimension and the exponent $3/4$ is not optimal; the true order is $\tau^2\log(1/\tau)$ per step. The purity version has neither defect. Only branches, with pure inputs, are covered: the entropy of the unconditioned iterate $(\Psi_{t/n})^n\rho$ for mixed ρ is not formalized.

Problems found in the brief. No errors in the statements. Gaps and implicit points:

- Proposition 2 shows that the integral (7) is a right inverse of $I-\tau L$. That it is $(I-\tau L)^{-1}$ also needs injectivity, which is not mentioned. In finite dimension it follows from eq. (11): for self-adjoint σ, $\sigma = \sum_x P_x(\sigma-\tau L\sigma)P_x$ over the spectral projectors of σ, so $(I-\tau L)\sigma = 0$ forces $\sigma = 0$.

- Proposition 13 appeals to "standard perturbation of an isolated eigenvalue" without a reference or a gap condition. Here it is a Davis–Kahan bound. Let $q \ge \frac{1}{2}$ be the largest eigenvalue of $\sigma = \Psi_\tau P$, $F = |\phi\rangle\langle\phi|$ its spectral projector and $u = V_\tau\psi$. Then $\sigma - qF$ is positive with trace $1-q$, so $\langle w|\sigma w\rangle \le (1-q)\|w\|^2$ for $w = u - \langle\phi|u\rangle\,\phi$. Comparing with $\sigma u = u + (\sigma - P_u)u$ gives $\|w\| \le 2\|\sigma - P_u\|$, and $\|F - P_u\| \le 3\|w\|$.

- Section 3 states DR-PROJ only as "the probabilities of the predicted and reconstructed state sequences tend to one". The remark on prediction and reconstruction adds that the predicted states converge. This is not an extra assumption: since the branch is within $2(1-w)$ of $\Psi^nP \to \Phi_tP$, the predicted states converge to Φ_tP anyway. The finite-dimensional proof still uses the clause; the proof in any dimension does not (Section 1.1).

- "Perfect reconstruction gives the converse correspondence" does not say how the reconstructed sequence relates to the predicted one. Injectivity and surjectivity of Φ_t on pure states need the reconstruction to invert Φ_t; this is what DR-PROJ assumes here.

- Composition for durations with irrational ratio, "take common refinements with a remaining interval whose duration tends to zero", is a gap. The remaining interval is harmless: with a common step τ the powers of Ψ_τ commute, so $\|\Psi_\tau^k(\Psi_\tau^j\rho-\rho)\|_1 \le j\|\Psi_\tau\rho-\rho\|_1 \approx j\tau\|L\rho\|_1 \to 0$ by the telescoping of eq. (28). The problem is the step size. DR-PROJ defines Φ_t only as $\lim_n(\Psi_{t/n})^n$. With t/s irrational and $\tau = (t+s)/N$, the block of $k = \lfloor t/\tau\rfloor$ steps is Ψ_τ^k, which is not of the form $(\Psi_{t/k})^k$, and DR-PROJ says nothing about its limit. Comparing $(\Psi_a)^k$ with $(\Psi_b)^k$ for nearby a, b needs the $o(\tau)$ in eq. (26) to be uniform along the states $\Psi_b^j\rho$, while eq. (26) holds for each ρ separately. Continuity at zero does not supply this. In finite dimension L is bounded, eq. (26) holds in operator norm, and telescoping gives $\|(\Psi_{t/n})^n - e^{tL}\| \le n\,o(t/n)\,e^{t\|L\|} \to 0$, so $\Phi_t = e^{tL}$ and composition follow at once (formalized, `IsProjectiveLimit.exists_eq_exp`). In infinite dimension Chernoff's product formula gives the same, uniformly on compact t-intervals, provided the closure of L generates a contraction semigroup; the brief's "single, possibly unbounded generator L" must be read to include this, or it must be assumed.

- "The phases of the implementing unitaries may be chosen" is the lifting step already noted for the brief on the Schrödinger equation; it is filled in the same way.

1.1 Hilbert spaces of any dimension

This subsection covers Definition 1, Propositions 2 and 13, eq. (20) and Theorem 18 in both directions, with an unbounded generator, and the remark "Unbounded generators" of Sec. 5.

Files.

- File: `briefs/UnitaryProjectiveEvolution/UnitaryProjectiveEvolution/InfiniteDimensional.lean`
- Library: `UnitaryProjectiveEvolution`, which now builds every module `UnitaryProjectiveEvolution.*`
- Imports: `ForPhyslib.QuantumMechanics.ForwardHamiltonian`, `ForPhyslib.QuantumMechanics.SpectralCutoff`, `ForPhyslib.QuantumMechanics.VonNeumannResolvent.Laplace`, `Mathlib.NumberTheory.Real.Irrational` (for `Irrational`), and `SchroedingerEvolution.InfiniteDimensional` (for `IsEvolution`, `DrUnit`, `DrEv` and `DrEv.drUnit`; it brings `StronglyContinuousUnitaryGroup` and `AffineOnDensityOperators`)
- For the remark "Unbounded generators": the file `UnboundedResolvent.lean` next to `InfiniteDimensional.lean`, importing `UnitaryProjectiveEvolution.InfiniteDimensional` and `ForPhyslib.QuantumMechanics.VonNeumannResolvent.Strong`
- New file `ForPhyslib/QuantumMechanics/VonNeumannResolvent/Strong.lean`: the resolvent of a strongly continuous unitary group as the Laplace integral taken pointwise on vectors, its agreement with the Laplace resolvent of a norm-continuous group, positivity, preservation of density operators, the derivative of $u \mapsto T_u \Psi_\tau \rho\, x$ at zero, the eigenbasis decomposition, the largest outcome on a pure state with a Lipschitz orbit, Lipschitz orbits on the domain of the generator, and complete positivity.
- New file `ForPhyslib/ProbabilisticTheory/CStarAlgebra/Stinespring/Kernel.lean`: a $k \times k$ matrix M of operators is positive exactly when $\sum_{ij} \langle x_i | M_{ij} x_j \rangle \geq 0$ for every $x \in E^k$. It extends PhyslibAlpha's block operator `blockMatrixMap`, an injective $*$-homomorphism, and an injective $*$-homomorphism of C*-algebras is an order embedding (Mathlib).
- New file `ForPhyslib/QuantumMechanics/VonNeumannResolvent/Laplace.lean`: the resolvent of a bounded generator in any dimension as the Laplace integral of the unitary channel, its inverse identities, complete positivity, preservation of density operators, its agreement with $(1 - \tau L)^{-1}$ in finite dimension, the eigenbasis decomposition, the second-order estimates and the largest outcome in trace norm, the exponential formula and the generator limit.
- New file `ForPhyslib/QuantumMechanics/SpectralCutoff.lean`: the cutoff groups $\int e^{it c_M(x)}\, d\mu_S(x)$ through PhyslibAlpha's bounded spectral integral, the bound $\|H_M\| \leq M$ on their generators, strong convergence as $M \to \infty$, and the spectral projectors $P_m = \mu_S[-m, m]$.
- Additions to `ForPhyslib/QuantumMechanics/TraceNorm.lean`: `traceENorm`, the trace norm $\mathrm{tr}\,|T| \in [0, \infty]$ of any operator, with its calculus; the trace-norm distance of two pure states

(`traceENorm_proj_sub_proj_le`); extension of trace-norm bounds from pure states to density operators (`traceENorm_apply_le_of_forall_proj`); trace-norm convergence of $U_a \rho U_a^\dagger$ when $U_a \to U$ strongly (`tendsto_traceENorm_conj`); density of the density operators supported in projections $P_a \to 1$ (`exists_isDensityOperator_mul_eq`); closedness of the pure states and of the density operators under trace-norm limits (`isPure_of_tendsto_traceENorm` and `isDensityOperator_of_tendsto_traceENorm`); uniqueness of trace-norm limits; and the affinity, contraction and telescoping bounds for iterates of affine maps.

- Addition to `ForPhyslib/QuantumMechanics/Stone.lean`: `StronglyContinuousUnitaryGroup.exists_expIntegral`, every strongly continuous unitary group is $t \mapsto \int e^{itx}\, d\mu_S(x)$ for a projection-valued measure μ_S on $\mathbb{R}$.
- New `ForMathlib` material: `ForMathlib/Analysis/CStarAlgebra/CompletelyPositiveMap.lean` (an integral of completely positive maps is completely positive, `CompletelyPositiveMap.integral`).

Setting. E is a complex Hilbert space of any dimension; separability is not used. Density operators (`IsDensityOperator`) and pure states are as in the section "Hilbert spaces of any dimension" of `briefs/SchroedingerEvolution/formalization-log.tex`, and DR-UNIT is `DrUnit` from there: $\Phi_t \rho = U_t \rho U_t^\dagger$ for a strongly continuous unitary group, whose generator may be unbounded. Limits are in the trace norm, through `traceENorm`. For a norm-continuous group $V_t = e^{-itK}$ with bounded generator, L, T_t and $\hbar = 1$ are as in finite dimension, and `laplaceResolvent` $V\ \tau = \int_0^\infty e^{-s} T_{\tau s}\, ds$ is Ψ_τ, a bounded operator on $\mathrm{End}(E)$. The definitions are those of finite dimension, with these changes:

- `IsProjectiveDecomposition` P: the index type is arbitrary and $\sum_a P_a = 1$ converges strongly; `IsStateAdaptiveProjective`, `IsOutcome`: the sums $\sum_a P_a \rho P_a$ converge strongly;
- `PerfectPrediction`: the final branch state converges in trace norm;
- `IsProjectiveLimit` $\Phi\ \Psi$: $(\Psi_{t/n})^n \rho \to \Phi_t \rho$ in trace norm, and the infinitesimal condition is eq. (26): there is a set D of density operators, dense in trace norm, such that $(\Psi_\tau \rho - \rho)/\tau$ converges in trace norm as $\tau \to 0^+$ for every $\rho \in D$.

Results.

Brief	Lean
Definition 1	`IsStateAdaptiveProjective`, `IsProjectiveDecomposition`
Eq. (7), Ψ_τ as a mixture of unitary channels	`laplaceResolvent`
Eqs. (8)–(9), $(I - \tau L)\Psi_\tau = I$	`integral_laplace_vonNeumannStep`, `laplaceResolvent_sub`
$\Psi_\tau(I - \tau L) = I$	`laplaceResolvent_vonNeumannStep`
Ψ_τ is affine, positive, trace preserving	`laplaceResolvent_nonneg`, `isDensityOperator_laplaceResolvent`
Proposition 2, Ψ_τ is completely positive	`laplaceResolventCP`, `coe_laplaceResolventCP`
Eq. (11), on an eigenbasis	`inner_vonNeumannGenerator_apply_eq_zero`

Brief	Lean
Eqs. (4), (12)	`exists_hilbertBasis_laplaceResolvent`: $\Psi_\tau\rho = \sum_i P_i\rho P_i$ with $P_i\rho P_i = \mu_i P_i$
Proposition 2	`isStateAdaptiveProjective_laplaceResolvent`
$\|L_K\| \le 2M/\hbar$ on trace-class operators	`traceENorm_vonNeumannGenerator_le`
Proposition 13, eq. (14), in trace norm	`traceENorm_laplaceResolvent_sub_unitaryChannel_le`: $\le 80\|K\|^2\tau^2$
Proposition 13, eq. (15), in trace norm	`exists_max_laplaceResolvent_proj`: $1 - q_\tau(P) \le 8\|K\|^2\tau^2$, $\|F_\tau(P) - T_\tau P\|_1 \le 64\|K\|^2\tau^2$
$(\Psi_\tau - I)/\tau = L\Psi_\tau \to L$, bounded K	`tendsto_inv_smul_laplaceResolvent_sub`, error $\le 4\|K\|^2\tau$
Eq. (20), bounded K	`tendsto_laplaceResolvent_iterate`, `traceENorm_laplaceResolvent_iterate_sub_le`: error $\le 80\|K\|^2t^2/n$
Eq. (22)	`IsCutoffLevel`; $M_\tau = \tau^{-1/4}$ satisfies it (`isCutoffLevel_cutoffLevel`)
$H_\tau = f_\tau(H)$, $\|H_\tau\| \le M_\tau$	`cutoffGroup`, `norm_cutoffGroup_generator_le`
$e^{tH_\tau/(i\hbar)} \to e^{tH/(i\hbar)}$ strongly	`tendsto_cutoffGroup_apply`
Eq. (23)	`cutoffResolvent`
Eq. (26), on the core of bounded spectral support	`infinitesimal_cutoffResolvent`, `laplaceResolvent_cutoffGroup_eq`, `cutoffGroup_mul_spectralCutoffProj`
The generator of family (23) is $[H,\rho]/(i\hbar)$, on the core	`cutoffResolvent_generator`, `expGroup_mul_spectralCutoffProj`, `exists_mem_domain_of_spectralCutoffProj`, `forwardHamiltonian_expGroup`
The core is dense	`exists_isDensityOperator_mul_eq`, `tendsto_spectralCutoffProj_apply`
Eqs. (24)–(25), hence eq. (20) for unbounded H	`tendsto_cutoffResolvent_iterate`, `traceENorm_sub_unitaryChannel_le_of_step`, `tendsto_traceENorm_conj`
Perfect prediction, unbounded H, total failure $O(t\tau M_\tau^2)$	`perfectPrediction_cutoffResolvent`
$U_t = e^{tH/(i\hbar)}$ for a spectral measure	`StronglyContinuousUnitaryGroup.exists_expIntegral`
"Replacing H_τ with $-H_\tau$"	`timeReversal`
DR-UNIT ⇒ DR-PROJ	`DrUnit.drProj`, from `isProjectiveLimit_cutoffResolvent` and `exists_isProjectiveLimit`
The branch has weight tending to one in the unconditioned output	`IsOutcome.nonneg`, `nonneg_iterate_sub_of_isBranch`, `traceENorm_iterate_sub_le_of_isBranch`: $\|\Psi^n Q_0 - Q_n\|_1 \le 2(1-w)$
Φ_t maps pure states to pure states	`IsProjectiveLimit.isPure`, `isPure_of_tendsto_traceENorm`
Φ_t maps pure states bijectively onto pure states	`DrProj.drEv`

Brief	Lean
Proposition 17 and eq. (27), phases	from the brief on the Schrödinger equation, `DrEv.drUnit` (the section "Hilbert spaces of any dimension" of `briefs/SchroedingerEvolution/formalization-log.tex`)
DR-PROJ ⇒ DR-UNIT	`DrProj.drUnit`
Theorem 18	`drUnit_iff_drProj`
Every finite repetition is affine; Φ_t is affine and keeps density operators	`AffineOnDensityOperators.iterate`, `IsProjectiveLimit.affine`, `IsProjectiveLimit.isDensityOperator`
Ψ_τ and Φ_t contract the trace norm	`AffineOnDensityOperators.traceENorm_sub_le`, `IsProjectiveLimit.traceENorm_sub_le`
Eq. (28), telescoping	`IsProjectiveLimit.traceENorm_sub_le_mul`, `AffineOnDensityOperators.traceENorm_iterate_sub_le`
Continuity at zero, extended by contractivity	`IsProjectiveLimit.tendsto_traceENorm_sub`, `IsProjectiveLimit.continuousWithinAt`
Eq. (29), composition for durations of rational ratio	`IsProjectiveLimit.map_add_mul`, `IsProjectiveLimit.map_add_of_not_irrational`
The evolution structure from DR-PROJ	`IsProjectiveLimit.isEvolution`, assuming $\Phi_0 = I$ and composition for durations of irrational ratio
Theorem 18 without `IsEvolution`	`drUnit_iff_drProj_and_map_zero_and_map_add`
Sec. 5, the resolvent of an unbounded generator, used directly	`resolvent`, `resolvent_eq_laplaceResolvent` (in `StronglyContinuousUnitaryGroup`)
It is positive and trace preserving	`resolvent_nonneg`, `isDensityOperator_resolvent`
It is completely positive	`exists_completelyPositiveMap_resolvent`, `resolventCP`, `cstarMatrix_nonneg_iff`
Proposition 2 for it, eq. (11) without a domain	`isStateAdaptiveProjective_resolvent`, `exists_hilbertBasis_resolvent`, `inner_apply_eq_of_resolvent_apply`, `hasDerivAt_conj_resolvent_apply`
Its resolvent exponential formula	`tendsto_resolvent_iterate`
Eq. (26) for it, on the core	`infinitesimal_resolvent`, `resolvent_eq_laplaceResolvent_cutoffGroup`, `resolvent_iterate_eq`
Perfect prediction on the domain of H, failure at most $8\|H\psi\|^2t^2/n$	`perfectPrediction_resolvent`, `perfectPrediction_resolvent_of_mem_domain`, `exists_max_resolvent_proj`, `norm_apply_sub_le_of_mem_domain`

Differences from the brief.

- Limits and the bounds of Proposition 13 are in the trace norm, as the brief's remark on prediction and reconstruction asks. `traceENorm` is $\operatorname{tr}|T|$ in $[0,\infty]$ for every operator, so convergence statements carry no trace-class witnesses; it is finite exactly on the trace class, where it is PhyslibAlpha's trace norm.

- In Definition 1 the projective decomposition has an arbitrary index type and its sums converge strongly. For a density operator ρ the sum $\sum_a P_a \rho P_a$ converges in trace norm anyway, so nothing is lost; in a separable space at most countably many P_a are nonzero.

- Ψ_τ is defined by the integral (7), which converges in operator norm on $\mathrm{End}(E)$ for every real τ. It is a two-sided inverse of $I - \tau L$: the brief's eqs. (8)–(9) give the right inverse, and since L commutes with every T_s the same integral is a left inverse. Positivity comes from the integral. That $\Psi_\tau \rho$ is trace class with trace one uses the integral again, now in the trace class $\mathcal{S}_1$, where $s \mapsto T_s \rho$ is continuous and the trace is a continuous functional. Interchanging a basis sum with the integral would need a countable basis.

- The decomposition of Proposition 2 uses the rank-one projectors P_i onto an eigenbasis of $\sigma = \Psi_\tau \rho$, not the spectral projectors that group equal eigenvalues. Both satisfy Definition 1. Eq. (11) becomes $\langle b_i | L\sigma\, b_i \rangle = 0$, so $P_i \rho P_i = \langle b_i | \rho\, b_i \rangle\, P_i = \mu_i P_i$, and "every P_a^P with $\lambda_a > 0$ has rank one" is automatic. For a pure state the largest eigenvalue is at least $1 - 8\|K\|^2\tau^2 > \frac{1}{2}$ and the eigenvalues sum to one, so it is simple and its eigenprojector is the brief's $F_\tau(P)$.

- Proposition 13 is proved through the finite-dimensional operator-norm argument: the Davis–Kahan bound $\|w\| \le 2\|\sigma - P_u\|$ for the component w of $u = V_\tau \psi$ orthogonal to the top eigenvector ϕ, then $\|P_\phi - P_u\|_1 \le 4\|w\|$ for unit vectors. For a pure state P, $\Psi_\tau P - q F_\tau(P)$ is positive with trace $1 - q$, which gives eq. (14) on pure states with constant 80. It extends to all density operators by summing the canonical expansion in $\mathcal{S}_1$. For $8\|K\|^2\tau^2 > \frac{1}{2}$ the bound is trivial, since $\|\Psi_\tau \rho - T_\tau \rho\|_1 \le 2$, so eq. (14) holds for every $\tau \ge 0$.

- The cutoff level is any M_τ with the two limits of eq. (22) (`IsCutoffLevel`), as in the brief. The brief's $M_\tau > 0$ is not assumed: $M_\tau \to \infty$ gives it for small τ, which is all the estimates use. The bounds are explicit at $\tau = t/n$: the accumulated error is at most $80\, t\tau M_\tau^2$, the failure probability at most $8\, t\tau M_\tau^2$, and the state error at most $64\, t\tau M_\tau^2$. The level $M_\tau = \tau^{-1/4}$, with $\tau M_\tau^2 = \sqrt{\tau}$, shows that such levels exist (`isCutoffLevel_cutoffLevel`), and `exists_isProjectiveLimit` uses it. H_τ is never formed as an operator of its own. The cutoff group $W_t = \int e^{itc_M(x)}\, d\mu_S(x)$, with $c_M(x) = \max(-M, \min(x, M))$, is a norm-continuous unitary group because $\|W_t - W_s\| \le M|t - s|$, and H_τ is its bounded generator, of norm at most M by the same Lipschitz bound. Up to the sign convention of physlib's `UnitaryOneParameterGroup`, it is the brief's $f_\tau(H)$. Strong convergence of the cutoff groups is dominated convergence in $\|(W_t - U_t)x\|^2 = \int |e^{itc_M(x)} - e^{itx}|^2\, d\mu_x$.

- In `IsProjectiveLimit`, eq. (26) does not name the generator: $(\Psi_\tau \rho - \rho)/\tau$ converges for ρ in a dense set D of density operators. D is the brief's core, the density operators with $P_m \rho = \rho$ for $P_m = \mu_S[-m, m]$ and some m. On such ρ, $W_t^{(M)} P_m = W_t^{(m)} P_m$ for $M \ge m$, so once $M_\tau \ge m$, $\Psi_\tau \rho$ is the resolvent of the fixed bounded group $W^{(m)}$, and the bounded case applies. The core is dense because $P_m \to 1$ strongly: $P_m \rho P_m / \operatorname{tr}(P_m \rho P_m) \to \rho$ in trace norm. The limit is $L_{K_m}\rho$, the commutator with the bounded part K_m of H. It is the brief's $[H, \rho]/(i\hbar)$ in the following sense (`cutoffResolvent_generator`): for the self-adjoint H with $U_t = e^{-itH}$ (Stone's theorem), ρ maps into the domain of H, and $L\rho\,\psi = -i(H\rho\psi - \rho H\psi)$ for ψ in the domain, which determines the bounded $L\rho$. On the range of P_m, $U_t = W_t^{(m)}$, so $H = K_m$ there; this uses that H is the forward Hamiltonian of U from the brief on the Schrödinger equation (`forwardHamiltonian_expGroup`). Then $\rho H \psi = \rho K_m \psi$ by symmetry of H.

- DR-UNIT ⇒ DR-PROJ needs no `IsEvolution`, as in finite dimension. Reconstruction uses the cutoff resolvents of the reversed group $t \mapsto U_{-t}$, which has its own spectral measure.

- DR-PROJ ⇒ DR-UNIT is proved over `IsEvolution` (`DrProj.drUnit`) and over two of its clauses only (`drUnit_iff_drProj_and_map_zero_and_map_add`). The finite-dimensional derivation of `IsEvolution`, through $\Phi_t = e^{tB}$, needs a finite basis of density operators; `IsProjectiveLimit.isEvolution` follows the brief instead. Each Φ_t is a trace-norm limit of affine maps that keep density operators, so it is one too, and contracts the trace norm (Simon's lemma, the brief's "extend to positive trace-preserving linear maps"). Eq. (28) is the telescoping bound $\|\Psi^n\rho - \rho\|_1 \le n\|\Psi\rho - \rho\|_1$ on the dense set of eq. (26), and contractivity extends continuity at 0^+ to every density operator. Eq. (29) is proved as $\Phi_{(p+q)u} = \Phi_{pu} \circ \Phi_{qu}$, along $n = (p+q)m$, which gives composition whenever s/t is rational. Two clauses remain assumptions: $\Phi_0 = I$, since `IsProjectiveLimit` defines Φ_t only for $t > 0$ (the brief obtains it from $\Psi_0 = I$), and $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for s/t irrational, the gap noted in the finite-dimensional part.

- The branch estimate is the finite-dimensional one, with traces in $[0, \infty]$: along a branch of probability w, $\Psi^n Q_0 - wQ_n$ is positive with trace $1 - w$, so it is $(1-w)R$ for a density operator R, and $\|\Psi^n Q_0 - Q_n\|_1 \le 2(1-w)$. In finite dimension the pure states are closed because they are the image of the compact unit sphere; here they are closed in $\mathcal{S}_1$, as the positive idempotents of trace norm one. Proposition 17 is Proposition 1 of the brief on the Schrödinger equation in any dimension, which needs affinity under all convex combinations; `IsStateAdaptiveProjective` and `IsEvolution` both assume it.

- The proof of DR-PROJ ⇒ DR-UNIT does not use the clause of `PerfectPrediction` that the final branch states converge. The bound $\|\Psi^n P - Q_n\|_1 \le 2(1 - w_n)$ and $\Psi^n P \to \Phi_t P$ already give $Q_n \to \Phi_t P$. The clause is kept so that the definitions match those of finite dimension.

- Complete positivity is Mathlib's `CompletelyPositiveMap`: $\Psi_\tau \otimes \mathrm{id}$ is positive on the $k \times k$ matrices of operators for every k. It holds because every unitary channel is a $*$-automorphism, and an integral of completely positive maps is completely positive, since the positive operators form a closed cone.

- The remark "Unbounded generators" is read as three claims about $\Psi_\tau = \int_0^\infty e^{-s} T_{\tau s}\, ds$ for the group U_t itself, with no cutoff. In operator norm $s \mapsto T_s$ is now only strongly continuous, so the integral is taken pointwise, $(\Psi_\tau \rho)x = \int_0^\infty e^{-s} T_{\tau s} \rho\, x\, ds$, a Bochner integral in E. On a positive trace-class ρ the integrand is still continuous in $\mathcal{S}_1$, so the trace argument above carries over.

  - "A completely positive trace-preserving map": formalized, together with Proposition 2 for every $\tau > 0$. `CompletelyPositiveMap.integral` needs an integral in operator norm, so complete positivity takes another route. A $k \times k$ matrix M of operators is positive exactly when $\sum_{ij} \langle x_i | M_{ij} x_j \rangle \ge 0$ for every $x \in E^k$. For the matrix of the $\Psi_\tau M_{ij}$ this sum is $\int_0^\infty e^{-s} \sum_{ij} \langle x_i | T_{\tau s} M_{ij}\, x_j \rangle\, ds$, and the integrand is nonnegative since $T_{\tau s}$ is completely positive.
  - Eq. (11) has no meaning when $L\sigma$ is not an operator. It is replaced by the derivative of $u \mapsto T_u \sigma\, x$ at 0, which is $(\sigma x - \rho x)/\tau$ by the shifted Laplace integral $T_u \sigma\, x = e^{u/\tau}(\sigma x - \int_0^{u/\tau} e^{-s} T_{\tau s} \rho\, x\, ds)$. For an eigenvector $\sigma b = \mu b$, $\langle b | (T_u \sigma - \sigma) U_u b \rangle = 0$ exactly, so $\langle b | T_u \sigma\, b \rangle - \mu$ is $O(u) \cdot o(1)$, and comparing derivatives gives $\langle b | \rho\, b \rangle = \mu$.

– "The resolvent exponential formula": $(\Psi_{t/n})^n\rho \to T_t\rho$ in trace norm for every density operator. On the core of eq. (26), Ψ_τ is the resolvent of the bounded group $W^{(m)}$ for every τ and keeps the core, since P_m commutes with $W^{(m)}$. So the bounded formula applies there, and trace-norm contraction extends it.

– "Only to ensure that the total probability ... tends to one ... including states outside the domain": prediction is perfect from every pure state P_ψ with ψ in the domain of H. The hypothesis used is a Lipschitz orbit, $\|U_u\psi-\psi\| \le C|u|$, which holds with $C = \|H\psi\|$ on the domain of the forward Hamiltonian. Testing $\Psi_\tau P_\psi$ on ψ gives a largest eigenvalue $q \ge 1-2C^2\tau^2$, since $1-|\langle\psi|U_u\psi\rangle|^2 \le \|U_u\psi-\psi\|^2$ and $\int_0^\infty s^2e^{-s}\,ds = 2$. Its eigenvector $\phi = q^{-1}\int_0^\infty e^{-s}\langle U_{\tau s}\psi|\phi\rangle\, U_{\tau s}\psi\,ds$ has a Lipschitz orbit with constant C/q. Along the branch of largest outcomes the constant stays below $2C$ once $8C^2t^2/n \le \frac{1}{2}$, so the branch probability is at least $1-8C^2t^2/n$. The final state converges by the branch estimate and the exponential formula.

So the untruncated family has every property of `IsProjectiveLimit` except `PerfectPrediction`, which asks for every pure state. Whether prediction fails outside the domain is not settled; the brief does not claim that it does.

Problems found in the brief. No errors. Gaps beyond those of the finite-dimensional part:

- Proposition 2 uses the spectral decomposition $\sigma = \sum_a \lambda_a P^\rho_a$ of $\sigma = \Psi_\tau(\rho)$. In infinite dimension this needs σ to be compact, which follows from σ being trace class; the brief does not show that it is. The decomposition may have infinitely many terms and includes the kernel of σ, and its sum converges strongly.

- The core of eq. (26) is asserted to be dense without proof. Density follows from $P_m \to 1$ strongly and trace-norm continuity of conjugation.

- Composition for durations of irrational ratio is the gap noted in the finite-dimensional part. Here no telescoping estimate closes it: L may be unbounded, and eq. (26) holds only for each ρ of a dense set, not uniformly.

- Eq. (25), "strong convergence implies convergence of the corresponding conjugations on every trace-class operator", is standard, but not obvious: for density operators it follows from the canonical expansion and the rank-one estimate $\|P_x - P_y\|_1 \le 2\|x-y\|$.